Lesson Plan

Teacher Name: Ms. Lubna Kakkengal and Mr. Dilan Devendra	**Class/Section:** Grade 5
Subject: TCD: Junior Designer Challenge	**Date: Week 13 (23 Nov to 27 Nov)**
Class Description: Grade 5 TCD class, Junior Designer Challenge Session 5 of 5 (1 hour 30 minutes). Students finalise their car logo model, check it against a print-ready checklist, export it, see how the slicer prepares it, and send it to the 3D printer. They compare the print with their plan.	
TA/LSA Role (if applicable):	

Differentiated Learning Groups (Data)			
Desert Fox	**Camel**	**Oryx**	**Falcon**

Lesson Overview		
Lesson Topic	Finalise and 3D Print	
Standard(s)	Technology: CAD to 3D print workflow; Math: 5.MD.C (volume), measurement and precision	
Learning Objective(s) (WALT): We are learning to prepare a 3D model so it prints well, and to check the print against our plan.		**Success Criteria (WILF):** I can check my model with the print-ready checklist. I can export my model as an STL file. I can explain what the slicer does, including layers, infill and supports. I can compare my print with my design and suggest one improvement.
Literacy Strategies	Print-ready checklist with picture prompts; vocabulary cards for STL, slicer, layer, infill and support.	
Assessment	Print-ready checklist; exported STL; print evaluation; live quiz on the TCD website (counts towards the Junior Designer Challenge certificate).	
Targeted Vocabulary	STL, export, slicer, layer, infill, support, nozzle, filament, print time	
Skills	Precision, evaluation, digital fabrication, patience	
National Identity	Link to Dubai's 3D Printing Strategy and the 3D printed Office of	

Integration	the Future in Dubai, showing how the UAE uses 3D printing in real buildings.
AI/Digital Learning	Tinkercad; slicer software (for example Ultimaker Cura) on the teacher's computer; hook video; live quiz on the TCD website. Presentation: Grade5_Week13_Presentation.pptx. Worksheet: Grade5_Week13_Worksheet.

Lesson Structure	
Starter (Hook) Engage and Ask (10 minutes)	Warm-up: Good Print, Bad Print (slide 4). Show the hook video and ask the hook question (slide 2). Share the learning goals (slide 3) and key words (slide 5).
I Do (20 minutes)	Go through the print-ready checklist using a model with problems (slide 6). Model exporting as STL (slide 7). Demonstrate the slicer on the projector (slide 8). Go through printer safety (slide 11).
We Do (10 minutes)	Partners check each other's model with the checklist and point out one fix.
You Do (40 minutes)	Students finalise and export their model (slides 12 and 10). Teacher slices and starts prints in batches; students evaluate prints as they finish (slides 9 and 13). Live quiz on tablets (slide 14).
Differentiated Instruction: Beginning / On / Above	**Beginning: Use the checklist with a partner and export with the step card.** **On: Fix all checklist problems and explain layers and infill.** **Above: Compare two infill settings in the slicer demo and explain which you would choose and why.**
Closure/ Reflection (10 minutes)	Prints that are ready are shown. Students compare with the reference. Exit ticket (slide 15). Prints not finished will be collected next week.

Inclusive Planning
Students receiving additional inclusive support (IEP or ILPs) or Gifted and Talented: EAL: picture checklist and vocabulary cards. SEN: partner checking and export step card. G&T: compare infill settings and justify the choice.
Inclusion / EAL Support

☑ Visual Supports
☑ Chunked Instructions
☐ Reduced Language Load
☑ Model / Worked Example
☐ Pre-teach Vocabulary
☐ TA/LSA Support
☐ Extra Processing Time
☐ Text Read by TA/LSA
☐ Flexible Grouping
☐ Assistive Technology
☐ Movement Break
☑ Scaffold / Sentence Starters
☐ Other:

G&T / Extension

☐ Extension / Challenge Task
☑ Higher-order Questioning
☑ Depth & Complexity Task
☐ Independent Inquiry
☐ Leadership / Peer Coaching
☑ Real-World Application
☐ Curriculum Compacting
☐ Above-grade-level Texts
☐ Student-led / Passion Project
☐ Open-Ended Task
☐ Other:

Assessment Adaptations

☑ Oral Response
☐ Reduced Questions
☑ Scaffolded Success Criteria
☐ Extra Time
☐ Alternative Outcome Format
☑ Teacher Conferencing/Live Feedback
☐ Quiet / Separate Space
☐ Other: